Plane Waves and Wave Propagation

PHYS505 – Electromagnetic Theory

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Plane waves in a nonconducting medium

- (semi-)infinite medium with no sources

$$\vec{\nabla} \cdot \vec{B} = 0 \qquad \vec{\nabla} \times \vec{E} + \partial_t \vec{B} = \vec{0}$$
$$\vec{\nabla} \cdot \vec{D} = 0 \qquad \vec{\nabla} \times \vec{H} - \partial_t \vec{D} = \vec{0}$$

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- Assuming harmonic time dependence, eg.

$$\vec{E}(\vec{x},t) = \vec{E}(\vec{x})e^{-i\omega t}, \quad \vec{B}(\vec{x},t) = \vec{B}(\vec{x})e^{-i\omega t}$$

$$\begin{aligned} \vec{\nabla}\cdot\vec{B}(\vec{x},t) &= 0 & \qquad \vec{\nabla}\times\vec{E}(\vec{x},t) + \partial_t\vec{B}(\vec{x},t) &= \vec{0} \\ \vec{\nabla}\cdot\vec{D}(\vec{x},t) &= 0 & \qquad \vec{\nabla}\times\vec{H}(\vec{x},t) - \partial_t\vec{D}(\vec{x},t) &= \vec{0} \end{aligned}$$

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- and substituting

$$\vec{\nabla}\begin{pmatrix}\cdot\\ \times\end{pmatrix}\left[\vec{E}(\vec{x})e^{-i\omega t}\right] = e^{-i\omega t}\vec{\nabla}\begin{pmatrix}\cdot\\ \times\end{pmatrix}\vec{E}(\vec{x})$$

$$\partial_t\left[\vec{E}(\vec{x})e^{-i\omega t}\right] = -i\omega e^{-i\omega t}\vec{E}(\vec{x})$$

$$e^{-i\omega t}\vec{\nabla}\cdot\vec{B}(\vec{x}) = 0 \qquad e^{-i\omega t}\vec{\nabla}\times\vec{E}(\vec{x}) - i\omega e^{-i\omega t}\vec{B}(\vec{x}) = \vec{0}$$

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- we get eqns for the (time-independent) field amplitudes

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- For uniform isotropic linear media

$$\vec{D} = \epsilon\,\vec{E}, \quad \vec{B} = \mu\,\vec{H}$$

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$$\vec{D} = \epsilon\,\vec{E}, \quad \vec{B} = \mu\,\vec{H}$$

- and we are left with $\vec{E}, \vec{B}$ only

$$\vec{\nabla}\cdot\vec{B}(\vec{x}) = 0 \qquad \vec{\nabla}\times\vec{E}(\vec{x}) - i\omega\vec{B}(\vec{x}) = \vec{0}$$

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- Taking the curl and substituting in each other

$$\vec{\nabla} \times \left(\vec{\nabla} \times \vec{E}\right) = \vec{\nabla}\left(\vec{\nabla} \cdot \vec{E}\right) - \nabla^2 \vec{E}$$

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$$\left(\nabla^2 + \mu\epsilon\omega^2\right)\vec{E} = \vec{0}$$
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- with plane-wave solutions of the form

$$\vec{E}(\vec{x}) = \vec{\mathcal{E}}e^{ik\hat{\mathbf{n}}\cdot\vec{x}}, \quad \vec{B}(\vec{x}) = \vec{\mathcal{B}}e^{ik\hat{\mathbf{n}}\cdot\vec{x}}, \quad k = \sqrt{\mu\epsilon}\,\omega$$

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- Phase velocity of the wave = velocity of (fixed-phase) wavefront

$$\vec{E}(\vec{x},t) = \vec{\mathcal{E}}e^{i(k\hat{\mathbf{n}}\cdot\vec{x}-\omega t)} = \vec{\mathcal{E}}e^{i\phi(\vec{x},t)}, \quad v = \frac{\omega}{k} = \frac{1}{\sqrt{\mu\epsilon}} = \frac{c}{n}$$

where $n = \sqrt{\frac{\mu}{\mu_0}\frac{\epsilon}{\epsilon_0}}$ is the *index of refraction*

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- If n is independent of ω, a wave packet made out of a superposition of plane waves, eg.

$$\vec{E}(\vec{x},t) = \vec{\mathcal{E}} \int_{-\infty}^{\infty} \frac{dk}{2\pi} F(k)\, e^{ik(\hat{\mathbf{z}}\cdot\vec{x}-vt)} = \vec{\mathcal{E}}\, f(z-vt)$$

  will hold its shape

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- Impedance Z of the medium defined through

$$\vec{\mathcal{H}} = \frac{1}{\mu}\vec{\mathcal{B}} = \frac{1}{Z}\hat{\mathbf{n}} \times \vec{\mathcal{E}}, \quad Z = \sqrt{\frac{\mu}{\epsilon}}, \quad Z_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} \simeq 376.73\,\Omega$$

Plane waves in a nonconducting medium

- Physical fields are the real parts of the complex quantities

$$\vec{E}(\vec{x},t) = \vec{\mathcal{E}} e^{i(k\hat{\mathbf{n}}\cdot\vec{x}-\omega t)}, \quad k = \sqrt{\mu\epsilon}\,\omega$$
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- $\vec{\mathcal{E}}, \vec{\mathcal{B}}$ are orthogonal to each other and the direction of propagation $\hat{\mathbf{n}}$
- For real $\hat{\mathbf{n}}$, fields are also in phase

$$\vec{\mathcal{E}} = E_0\,\hat{\epsilon}_1, \quad \vec{\mathcal{B}} = \sqrt{\mu\epsilon}\,E_0\,\hat{\epsilon}_2, \quad \hat{\epsilon}_1 \times \hat{\epsilon}_2 = \hat{\mathbf{n}}$$

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- The transverse wave represents time-averaged flux of energy given by the real part of $\vec{S}$

$$\vec{S} = \frac{1}{2}\vec{E} \times \vec{H}^* = \frac{Z}{2}|E_0|^2\,\hat{\mathbf{n}} = v\,u\,\hat{\mathbf{n}}$$
$$u = \frac{1}{4}\left(\epsilon|\vec{E}|^2 + \frac{1}{\mu}|\vec{B}|^2\right) = \frac{\epsilon}{2}|E_0|^2$$

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- $\vec{\mathcal{E}}, \vec{\mathcal{B}}$ are orthogonal to each other and the direction of propagation $\hat{\mathbf{n}}$
- For complex $\hat{\mathbf{n}}$, we have an *inhomogeneous plane wave*

$$\hat{\mathbf{n}} = \vec{n}_R + i\vec{n}_I, \quad \hat{\mathbf{n}}\cdot\hat{\mathbf{n}} = n_R^2 - n_I^2 = 1, \quad \vec{n}_r \cdot \vec{n}_I = 0$$
$$e^{i(k\hat{\mathbf{n}}\cdot\vec{x}-\omega t)} = e^{-k\vec{n}_I\cdot\vec{x}} e^{i(k\vec{n}_R\cdot\vec{x}-\omega t)}$$

Polarization

- General solution (with real $\hat{\epsilon}_1, \hat{\epsilon}_2$)

$$\vec{E}(\vec{x}, t) = (E_1\,\hat{\epsilon}_1 + E_2\,\hat{\epsilon}_2)\, e^{i(\vec{k}\cdot\vec{x}-\omega t)}$$

$$\vec{B} = \frac{1}{\omega}\vec{k}\times\vec{E}, \quad \vec{k} = k\,\hat{\mathbf{n}}, \quad \hat{\epsilon}_1\cdot\hat{\epsilon}_2 = 0, \quad \hat{\epsilon}_1\times\hat{\epsilon}_2 = \hat{\mathbf{n}}$$

  referred to as *elliptical polarization*

- *Linear polarization* when the transverse components E_1, E_2 are in phase

$$\vec{E}(\vec{x}, t) = \sqrt{|E_1|^2 + |E_2|^2}\,(\cos\theta\,\hat{\epsilon}_1 + \sin\theta\,\hat{\epsilon}_2)\, e^{i(\vec{k}\cdot\vec{x}-\omega t)}$$

- *Circular polarization* when they have the same magnitude but a phase difference of $\pi/2$

$$\vec{E}(\vec{x}, t) = E_0\,(\hat{\epsilon}_1 \pm i\hat{\epsilon}_2)\, e^{i(\vec{k}\cdot\vec{x}-\omega t)}$$

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Polarization

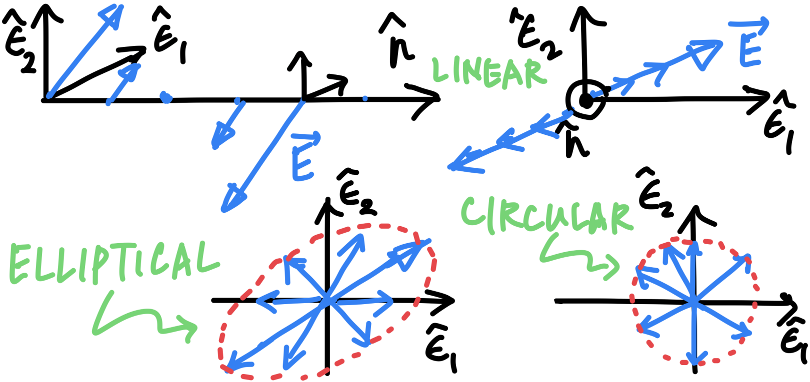
- General solution can also be written in the circular polarization basis as

$$\vec{E}(\vec{x},t) = (E_+\,\hat{\epsilon}_+ + E_-\,\hat{\epsilon}_-)\,e^{i(\vec{k}\cdot\vec{x}-\omega t)} \qquad \hat{\epsilon}_\pm = \frac{1}{\sqrt{2}}(\hat{\epsilon}_1 \pm i\,\hat{\epsilon}_2)$$

$$\hat{\epsilon}^*_\pm \cdot \hat{\epsilon}_\mp = \hat{\epsilon}^*_\pm \cdot \hat{\epsilon}_3 = 0 \qquad \hat{\epsilon}^*_\pm \cdot \hat{\epsilon}_\pm = 1$$

$$\hat{\epsilon}_- \times \hat{\epsilon}_+ = i\,\hat{\mathbf{n}}$$

Polarization



Polarization – Stokes parameters

$$\vec{E}(\vec{x},t) = \left(E_1\,\hat{\epsilon}_1 + E_2\,\hat{\epsilon}_2\right) e^{i(\vec{k}\cdot\vec{x}-\omega t)}$$

Stokes parameters involve quantities that can be determined by intensity measurements only

$$\begin{aligned}
E_1 &= a_1\, e^{i\delta_1}, \quad E_2 = a_2\, e^{i\delta_2} \\
s_0 &= \left|\hat{\epsilon}_1\cdot\vec{E}\right|^2 + \left|\hat{\epsilon}_2\cdot\vec{E}\right|^2 = a_1^2 + a_2^2 \\
s_1 &= \left|\hat{\epsilon}_1\cdot\vec{E}\right|^2 - \left|\hat{\epsilon}_2\cdot\vec{E}\right|^2 = a_1^2 - a_2^2 \\
s_2 &= 2\mathrm{Re}\left[\left(\hat{\epsilon}_1\cdot\vec{E}\right)^*\left(\hat{\epsilon}_2\cdot\vec{E}\right)\right] = 2a_1a_2\cos\left(\delta_2-\delta_1\right) \\
s_3 &= 2\mathrm{Im}\left[\left(\hat{\epsilon}_1\cdot\vec{E}\right)^*\left(\hat{\epsilon}_2\cdot\vec{E}\right)\right] = 2a_1a_2\sin\left(\delta_2-\delta_1\right) \\
s_0^2 &= s_1^2 + s_2^2 + s_3^2
\end{aligned}$$

Polarization

$$\vec{E}(\vec{x}, t) = (E_+ \hat{\epsilon}_+ + E_- \hat{\epsilon}_-) e^{i(\vec{k}\cdot\vec{x}-\omega t)}$$

Stokes parameters involve quantities that can be determined by intensity measurements only

$$E_+ = a_+ e^{i\delta_+}, \quad E_- = a_- e^{i\delta_-}$$

$$s_0 = \left|\hat{\epsilon}_+^* \cdot \vec{E}\right|^2 + \left|\hat{\epsilon}_-^* \cdot \vec{E}\right|^2 = a_+^2 + a_-^2$$

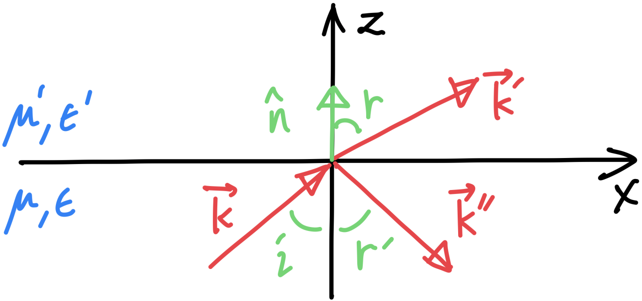
$$s_1 = 2\mathrm{Re}\left[\left(\hat{\epsilon}_+^* \cdot \vec{E}\right)^* \left(\hat{\epsilon}_-^* \cdot \vec{E}\right)\right] = 2a_+a_- \cos(\delta_- - \delta_+)$$

$$s_2 = 2\mathrm{Im}\left[\left(\hat{\epsilon}_+^* \cdot \vec{E}\right)^* \left(\hat{\epsilon}_-^* \cdot \vec{E}\right)\right] = 2a_+a_- \sin(\delta_- - \delta_+)$$

$$s_3 = \left|\hat{\epsilon}_+^* \cdot \vec{E}\right|^2 - \left|\hat{\epsilon}_-^* \cdot \vec{E}\right|^2 = a_+^2 - a_-^2$$

$$s_0^2 = s_1^2 + s_2^2 + s_3^2$$

Interface between media with indices n, n' in the $x - y$ plane



Incident	refracted	reflected
$\vec{E} = \vec{E}_0 e^{i(\vec{k}\cdot\vec{x}-\omega t)}$	$\vec{E}' = \vec{E}_0' e^{i(\vec{k}'\cdot\vec{x}-\omega t)}$	$\vec{E}'' = \vec{E}_0'' e^{i(\vec{k}''\cdot\vec{x}-\omega t)}$
$\vec{B} = \sqrt{\mu\epsilon}\,\hat{\mathbf{k}} \times \vec{E}$	$\vec{B}' = \sqrt{\mu'\epsilon'}\,\hat{\mathbf{k}}' \times \vec{E}$	$\vec{B}'' = \sqrt{\mu\epsilon}\,\hat{\mathbf{k}}'' \times \vec{E}''$

Dielectric interfaces – reflection & refraction

Interface between media with indices n, n' in the $x - y$ plane

Incident	refracted	reflected
$\vec{E} = \vec{E}_0 e^{i(\vec{k}\cdot\vec{x}-\omega t)}$	$\vec{E}' = \vec{E}'_0 e^{i(\vec{k}'\cdot\vec{x}-\omega t)}$	$\vec{E}'' = \vec{E}''_0 e^{i(\vec{k}''\cdot\vec{x}-\omega t)}$
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Matching the phases at the boundary

$$\vec{k}\cdot\vec{x} = \vec{k}'\cdot\vec{x} = \vec{k}''\cdot\vec{x}, \quad z = 0$$

$$\rightarrow \left(\vec{k} - \vec{k}'\right) \times \hat{\mathbf{n}} = \left(\vec{k} - \vec{k}''\right) \times \hat{\mathbf{n}} = \left(\vec{k}' - \vec{k}''\right) \times \hat{\mathbf{n}} = 0$$

Dielectric interfaces – reflection & refraction

Interface between media with indices n, n' in the $x - y$ plane

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Matching the phases at the boundary

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all wavevectors must lie in the same plane (called the *plane of incidence*) and satisy

$$k \sin i = k' \sin r = k \sin r' \rightarrow i = r', \quad \frac{\sin i}{\sin r} = \frac{k'}{k} = \sqrt{\frac{\mu'\epsilon'}{\mu\epsilon}} = \frac{n'}{n}$$

Interface between media with indices n, n' in the $x - y$ plane

Incident	refracted	reflected
$\vec{E} = \vec{E}_0 e^{i(\vec{k}\cdot\vec{x}-\omega t)}$	$\vec{E}' = \vec{E}_0' e^{i(\vec{k}'\cdot\vec{x}-\omega t)}$	$\vec{E}'' = \vec{E}_0'' e^{i(\vec{k}''\cdot\vec{x}-\omega t)}$
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Matching $\vec{D}_z, \vec{B}_z, \vec{E}_{xy}, \vec{H}_{xy}$ at the boundary ($z = 0$)

Dielectric interfaces – reflection & refraction

Interface between media with indices n, n' in the $x - y$ plane

Incident	refracted	reflected
$\vec{E} = \vec{E}_0 e^{i(\vec{k}\cdot\vec{x}-\omega t)}$	$\vec{E}' = \vec{E}'_0 e^{i(\vec{k}'\cdot\vec{x}-\omega t)}$	$\vec{E}'' = \vec{E}''_0 e^{i(\vec{k}''\cdot\vec{x}-\omega t)}$
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Matching $\vec{D}_z, \vec{B}_z, \vec{E}_{xy}, \vec{H}_{xy}$ at the boundary $(z = 0)$

$$\left[\epsilon\left(\vec{E}_0 + \vec{E}''_0\right) - \epsilon' \vec{E}'_0\right] \cdot \hat{\mathbf{n}} = 0$$

$$\left[\vec{k} \times \vec{E}_0 + \vec{k}'' \times \vec{E}''_0 - \vec{k}' \times \vec{E}'_0\right] \cdot \hat{\mathbf{n}} = 0$$

$$\left(\vec{E}_0 + \vec{E}''_0 - \vec{E}'_0\right) \times \hat{\mathbf{n}} = 0$$

$$\left[\frac{1}{\mu}\left(\vec{k} \times \vec{E}_0 + \vec{k}'' \times \vec{E}''_0\right) - \frac{1}{\mu'}\left(\vec{k}' \times \vec{E}'_0\right)\right] \times \hat{\mathbf{n}} = 0$$

Dielectric interfaces – reflection & refraction

Interface between media with indices n, n' in the $x - y$ plane

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$\vec{E} = \vec{E}_0 e^{i(\vec{k}\cdot\vec{x}-\omega t)}$	$\vec{E}' = \vec{E}_0' e^{i(\vec{k}'\cdot\vec{x}-\omega t)}$	$\vec{E}'' = \vec{E}_0'' e^{i(\vec{k}''\cdot\vec{x}-\omega t)}$
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For $\vec{E} \perp$ plane of incidence

Dielectric interfaces – reflection & refraction

Interface between media with indices n, n' in the $x - y$ plane

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$\vec{E} = \vec{E}_0 e^{i(\vec{k}\cdot\vec{x}-\omega t)}$	$\vec{E}' = \vec{E}'_0 e^{i(\vec{k}'\cdot\vec{x}-\omega t)}$	$\vec{E}'' = \vec{E}''_0 e^{i(\vec{k}''\cdot\vec{x}-\omega t)}$
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For $\vec{E} \perp$ plane of incidence

$$E_0 + E''_0 - E'_0 = 0$$

$$\sqrt{\frac{\epsilon}{\mu}}\left(E_0 - E''_0\right)\cos i - \sqrt{\frac{\epsilon'}{\mu'}}E'_0 \cos r = 0$$

Dielectric interfaces – reflection & refraction

Interface between media with indices n, n' in the $x - y$ plane

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$\vec{E} = \vec{E}_0 e^{i(\vec{k}\cdot\vec{x}-\omega t)}$	$\vec{E}' = \vec{E}'_0 e^{i(\vec{k}'\cdot\vec{x}-\omega t)}$	$\vec{E}'' = \vec{E}''_0 e^{i(\vec{k}''\cdot\vec{x}-\omega t)}$
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For $\vec{E} \perp$ plane of incidence

$$\frac{E'_0}{E_0} = \frac{2n\cos i}{n\cos i + \frac{\mu}{\mu'}\sqrt{n'^2 - (n\sin i)^2}}$$

$$\frac{E''_0}{E_0} = \frac{n\cos i - \frac{\mu}{\mu'}\sqrt{n'^2 - (n\sin i)^2}}{n\cos i + \frac{\mu}{\mu'}\sqrt{n'^2 - (n\sin i)^2}}$$

Dielectric interfaces – reflection & refraction

Interface between media with indices n, n' in the $x - y$ plane

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$\vec{E} = \vec{E}_0 e^{i(\vec{k}\cdot\vec{x}-\omega t)}$	$\vec{E}' = \vec{E}_0' e^{i(\vec{k}'\cdot\vec{x}-\omega t)}$	$\vec{E}'' = \vec{E}_0'' e^{i(\vec{k}''\cdot\vec{x}-\omega t)}$
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For $\vec{E} \parallel$ plane of incidence

Dielectric interfaces – reflection & refraction

Interface between media with indices n, n' in the $x - y$ plane

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$\vec{E} = \vec{E}_0 e^{i(\vec{k}\cdot\vec{x}-\omega t)}$	$\vec{E}' = \vec{E}'_0 e^{i(\vec{k}'\cdot\vec{x}-\omega t)}$	$\vec{E}'' = \vec{E}''_0 e^{i(\vec{k}''\cdot\vec{x}-\omega t)}$
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For $\vec{E} \parallel$ plane of incidence

$$\cos i \left(E_0 - E''_0\right) - \cos r E'_0 = 0$$

$$\sqrt{\frac{\epsilon}{\mu}}\left(E_0 + E''_0\right) - \sqrt{\frac{\epsilon'}{\mu'}} E'_0 = 0$$

Dielectric interfaces – reflection & refraction

Interface between media with indices n, n' in the $x - y$ plane

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For $\vec{E} \parallel$ plane of incidence

$$\frac{E_0'}{E_0} = \frac{2nn'\cos i}{\frac{\mu}{\mu'}n'^2\cos i + n\sqrt{n'^2 - (n\sin i)^2}}$$

$$\frac{E_0''}{E_0} = \frac{\frac{\mu}{\mu'}n'^2\cos i - n\sqrt{n'^2 - (n\sin i)^2}}{\frac{\mu}{\mu'}n'^2\cos i + n\sqrt{n'^2 - (n\sin i)^2}}$$

Dielectric interfaces – reflection & refraction

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For normal incidence ($i = 0$)

$$\frac{E'_0}{E_0} = \frac{2}{\sqrt{\frac{\mu}{\mu'}\frac{\epsilon'}{\epsilon}} + 1}$$

$$\frac{E''_0}{E_0} = \pm\frac{\sqrt{\frac{\mu}{\mu'}\frac{\epsilon'}{\epsilon}} - 1}{\sqrt{\frac{\mu}{\mu'}\frac{\epsilon'}{\epsilon}} + 1}$$

+/- sign for $\parallel$ / $\perp$ polarization

Dielectric interfaces – reflection & refraction

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For normal incidence ($i = 0$)

$$\frac{E_0'}{E_0} = \frac{2}{\sqrt{\frac{\mu}{\mu'}\frac{\epsilon'}{\epsilon}} + 1} \rightarrow \frac{2n}{n' + n}, \quad \mu' = \mu$$

$$\frac{E_0''}{E_0} = \pm\frac{\sqrt{\frac{\mu}{\mu'}\frac{\epsilon'}{\epsilon}} - 1}{\sqrt{\frac{\mu}{\mu'}\frac{\epsilon'}{\epsilon}} + 1} \rightarrow \pm\frac{n' - n}{n' + n}$$

+/- sign for $\parallel$ / $\perp$ polarization

Polarization by reflection

- For $\vec{E} \parallel$ plane of incidence

$$\frac{E_0''}{E_0} = \frac{\frac{\mu}{\mu'} n'^2 \cos i - n\sqrt{n'^2 - (n \sin i)^2}}{\frac{\mu}{\mu'} n'^2 \cos i + n\sqrt{n'^2 - (n \sin i)^2}}$$

- with $\mu' = \mu$, no reflected wave when angle of incidence is Brewster's angle

$$i = i_b = \tan^{-1}\left(\frac{n'}{n}\right) \rightarrow E_0'' = 0$$

- For a wave with mixed polarization incident at Brewster's angle, the reflection is purely *plane-polarized* with polarization $\perp$ plane of incidence
- Reflection at other angles typically dominated by the $\perp$ component

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Total internal reflection

- For $n' < n$, no refracted wave when angle of incidence is the critical angle
$$i_0 = \sin^{-1}\left(\frac{n'}{n}\right)$$
referred to as *total internal reflection*
- For $i > i_0$, there is an *evanescent wave* on the other side with exponentially decaying amplitude away from the interface and carrying no energy
- and a phase shift for the reflected wave
- equivalent, for a wave with finite transverse size, to a lateral displacement (Goos-Hänchen effect)

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Atomic contribution to dielectric constant

- Equation of motion for an electron in a region with a harmonic (in space) potential and an electric field

$$m\left(\frac{d^2}{dt^2}\vec{x}+\gamma\frac{d}{dt}\vec{x}+\omega_0^2\vec{x}\right)=-e\vec{E}(\vec{x},t)$$

- For harmonic time-dependence of the electric field, $\vec{E}(\vec{x},t)=\vec{E}e^{-i\omega t}$, the corresponding dipole moment due to the electron (assumed to be oscillating at a small amplitude)

$$\vec{p}\simeq -e\vec{x}=\frac{e^2}{m}\left(\omega_0^2-\omega^2-i\omega\gamma\right)^{-1}\vec{E}$$

- and for molecules with Z electrons in multiple states j

$$\frac{\epsilon(\omega)}{\epsilon_0}=1+\frac{Ne^2}{\epsilon_0 m}\sum_j\frac{f_j}{\omega_j^2-\omega^2-i\omega\gamma_j},\quad \sum_j f_j=Z$$

where N is the number density of molecules

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Dispersion in dielectrics

$$\frac{\epsilon(\omega)}{\epsilon_0} = 1 + \frac{Ne^2}{\epsilon_0 m} \sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\omega\gamma_j}, \quad \sum_j f_k = Z$$

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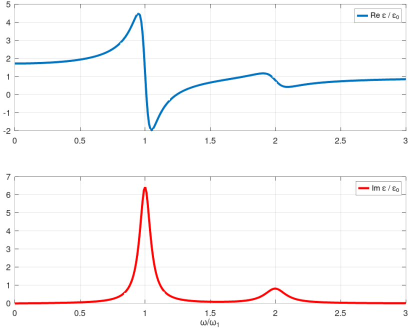
- Resonant frequencies ω_j (related to the energy levels), oscillator strengths f_j (related to the occupancy of states) and damping constants γ_j come from outside classical electrodynamics
- Typically $\gamma_j \ll \omega_j$, so ϵ is mostly real
- At low enough frequency when ω is less than all the ω_j, all the terms are +ve and $\epsilon/\epsilon_0 > 1$
- At high enough frequency when ω is greater than all ω_j, all the terms are -ve and $\epsilon/\epsilon_0 < 1$

Dispersion in dielectrics

$$\frac{\epsilon(\omega)}{\epsilon_0} = 1 + \frac{Ne^2}{\epsilon_0 m} \sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\omega\gamma_j}, \quad \sum_j f_k = Z$$

- For $\omega \approx \omega_j$, *resonant absorption* (large α) and *anomalous dispersion* ($d/d\omega Re[\epsilon] < 0$)



$$k = \beta + i\frac{\alpha}{2}, \quad \left|e^{ikz}\right|^2 \propto e^{-\alpha z}, \quad \frac{\omega}{k} = \frac{c}{n}$$

$$\beta^2 - \frac{\alpha^2}{4} = \left(\frac{\omega}{c}\right)^2 \mathrm{Re}\left[\frac{\epsilon}{\epsilon_0}\right]$$

$$\beta\alpha = \left(\frac{\omega}{c}\right)^2 \mathrm{Im}\left[\frac{\epsilon}{\epsilon_0}\right]$$

Dispersion in conductors

$$\frac{\epsilon(\omega)}{\epsilon_0} = 1 + \frac{Ne^2}{\epsilon_0 m}\sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\omega\gamma_j}, \quad \sum_j f_k = Z$$

- For conductors, fraction f_0 of electrons are free ($\omega_0 = 0$),

$$\epsilon(\omega) = \epsilon_b(\omega) + i\,\frac{\sigma}{\omega}, \quad \sigma = \frac{f_0 Ne^2}{m(\gamma_0 - i\omega)}$$

where ϵ_b is the contribution from bound electrons and

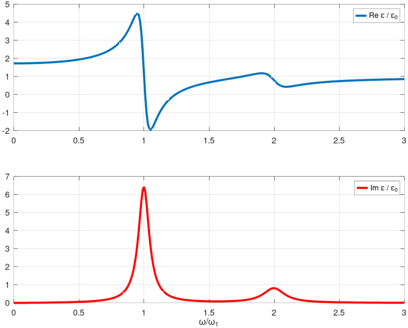
$$\vec{J} = \sigma\vec{E}, \quad \vec{D} = \epsilon_b\vec{E} \Rightarrow \vec{\nabla}\times\vec{H} = \vec{J} + \partial_t\vec{D} \to -i\omega\left(\epsilon_b + i\frac{\sigma}{\omega}\right)\vec{E}$$

For Cu, $\gamma_0 \simeq 4\times 10^{13}$ Hz

Dispersion in plasmas

$$\frac{\epsilon(\omega)}{\epsilon_0} = 1 + \frac{Ne^2}{\epsilon_0 m}\sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\omega\gamma_j}, \quad \sum_j f_k = Z$$

- For $\omega \gg \omega_j$ in dielectrics and also at lower frequencies in plasmas



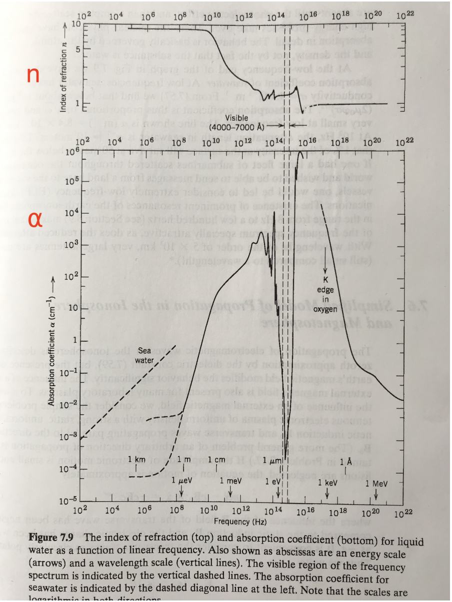
$$\frac{\epsilon}{\epsilon_0} \simeq 1 - \frac{\omega_p^2}{\omega^2}$$

$$\omega_p^2 = \frac{NZe^2}{\epsilon_0 m}$$

$$ck = \sqrt{\omega^2 - \omega_p^2}$$

$$\alpha_{pls} = \frac{2\omega_p}{c}$$

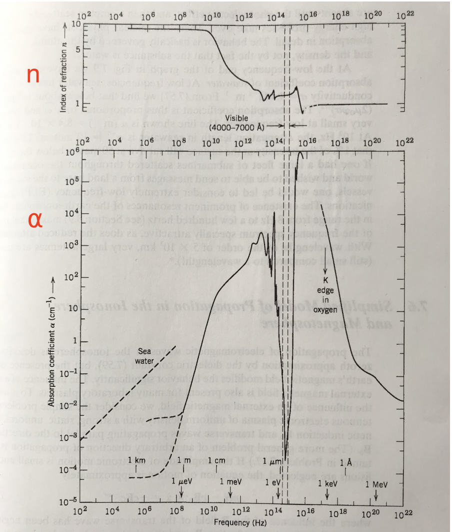
Water



$$n(\omega) = \mathrm{Re}\left[\sqrt{\frac{\mu}{\mu_0}\frac{\epsilon}{\epsilon_0}}\right]$$

$$\alpha(\omega) = 2\omega\ \mathrm{Im}\left[\sqrt{\mu\epsilon}\right]$$

Water



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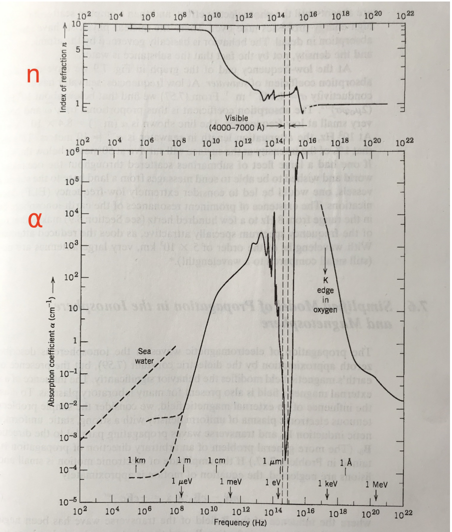

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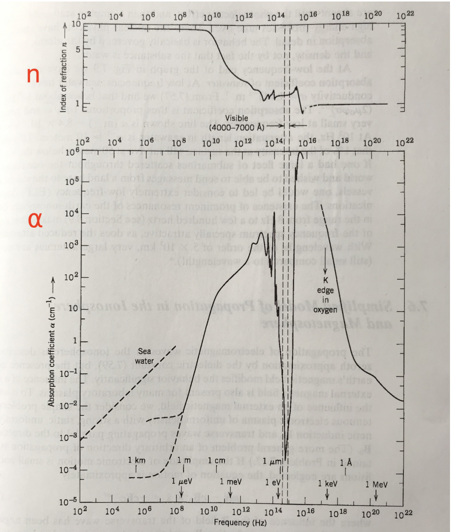
- $n(\omega \approx 0) \simeq 9$ due to the permanent dipole moment of H2O molecules
- $n \simeq 1.34$ and flat in the visible range

Water



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- α very small below 100 MHz

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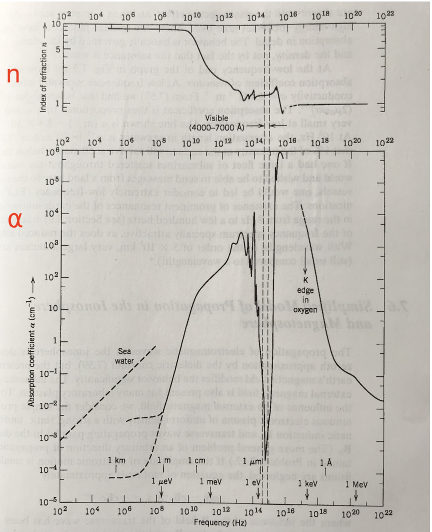

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- $1/\alpha \approx 1$ m at 300 MHz

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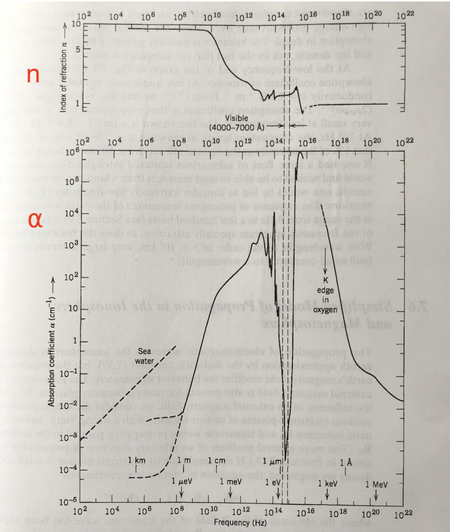

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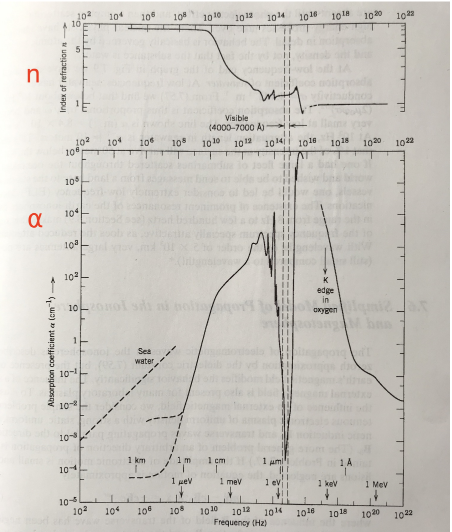
- α very small below 100 MHz
- $1/\alpha \approx 1$ m at 300 MHz
- $1/\alpha \approx 2$ cm at 2.5 GHz

Water



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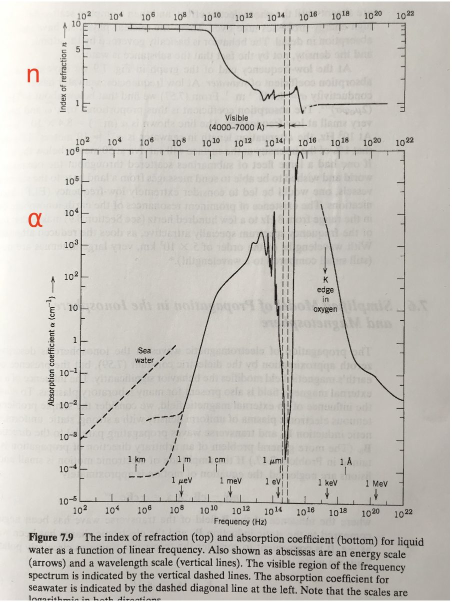

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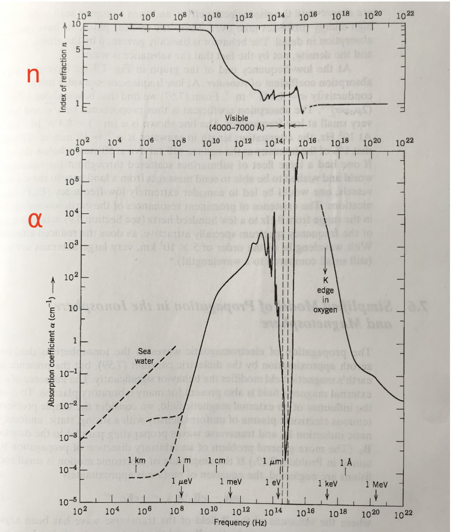
- absorption window in visible light



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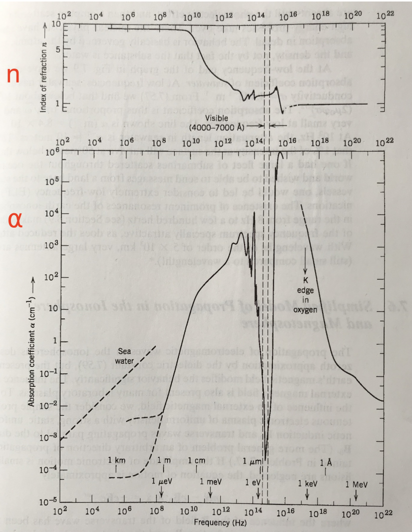
- large α at far/mid-infrared from rotation/vibration modes
- absorption window in visible light
- excited electronic states in uv

Water



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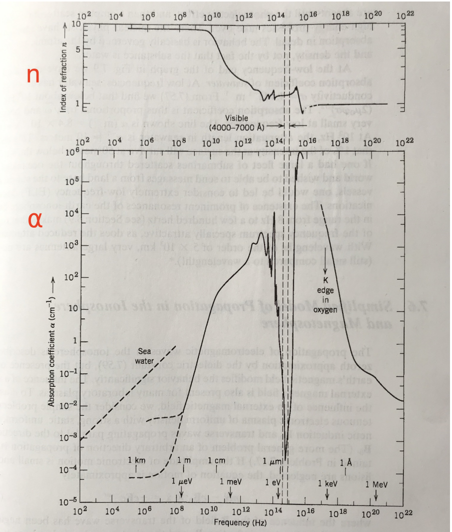
- peak $\alpha \simeq 10^8$/m at plasma frequency $\omega_p \simeq \pi \times 10^{16}$ Hz corresponding to collective excitation of electrons

Water



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- higher energy processes (photoelectric effect, Compton scattering, etc.) above

Ionosphere and magnetosphere

- Propagation of EM waves in the ionosphere (low-density plasma) affected by the external magnetic field
- Equation of motion for electrons with transverse wave propagating along the $\vec{B}_0$ direction

$$m\frac{d^2}{dt^2}\vec{x} - e\vec{B}_0 \times \frac{d}{dt}\vec{x} = -e\vec{E}e^{-i\omega t}$$

- leads to different behavior for the two circular polarization states

$$\vec{E} = \left(\hat{\epsilon}_1 \pm i\,\hat{\epsilon}_2\right) E$$

$$\vec{x} = \frac{e}{m}\frac{1}{\omega\left(\omega \pm \omega_B\right)}\vec{E}, \quad \omega_B = \frac{eB_0}{m}$$

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- For the ionosphere, $N \approx 10^{10} - 10^{12}/\mathrm{m}^3$ and $\omega_p \approx 6 - 60$ MHz and the precession frequency $\omega_B \approx 6$ MHz
- Wide frequency intervals where only one helicity can propagate with the other helicity state reflected
- Propagation further complicated by density gradient

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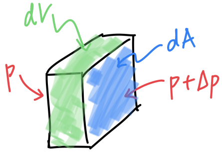

$\vec{f}\, dV = \Delta p\, dA \Rightarrow \vec{f} = \vec{\nabla} p$

Force density

- mechanical, due to pressure gradient

$$\vec{f} = \vec{\nabla} p$$

- magnetic, due to external B-field

$$\vec{f} = \vec{J} \times \vec{B}$$

- Mass flux for flow velocity field $\vec{v}$: $\vec{J}_m = \rho\, \vec{v}$
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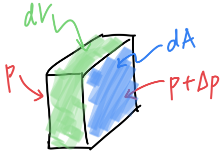
Notes on fluid dynamics



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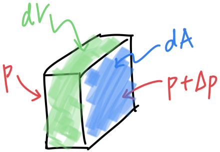
Notes on fluid dynamics



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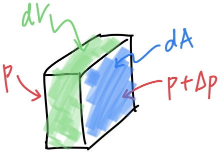

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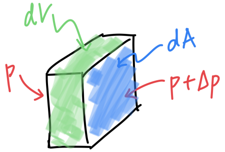

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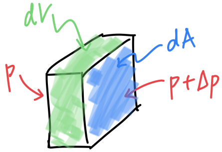

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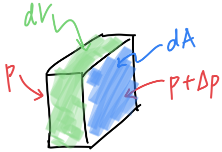

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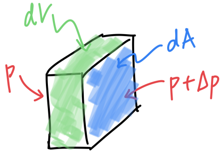
Notes on fluid dynamics



$\vec{f}\, dV = \Delta p \, dA \Rightarrow \vec{f} = \vec{\nabla} p$

Force density

- mechanical, due to pressure gradient
$$\vec{f} = \vec{\nabla} p$$
- magnetic, due to external B-field
$$\vec{f} = \vec{J} \times \vec{B}$$

- Mass flux for flow velocity field $\vec{v}$: $\vec{J}_m = \rho\, \vec{v}$
- Continuity eqn (mass conservation): $\partial_t \rho + \vec{\nabla} \cdot (\rho\, \vec{v}) = 0$
- Equation of state (adiabatic gas law): $p = K\, \rho^\gamma$
- Speed of sound $s^2 = \frac{\partial p}{\partial \rho} = \gamma\, K\, \rho^{\gamma-1} = \gamma p / \rho$
- Mechanical force density: $\vec{f} = \vec{\nabla} p = \gamma\, K\, \rho^{\gamma-1}\, \vec{\nabla} \rho = s^2\, \vec{\nabla} \rho$
- Momentum density, eqn of motion:
$\vec{p} = \rho\, \vec{v}$, $\vec{f} = \frac{d}{dt} \vec{p} = \rho \left(\partial_t + \vec{v} \cdot \vec{\nabla} \right) \vec{v}$

Magnetohydrodynamic waves

- Magnetohydrodynamics (MHD) refers to system of equations describing the motion of conducting fluid in terms of hydrodynamic variables (density ρ, velocity $\vec{v}$, pressure p) acted on by EM and gravitational forces

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$$\vec{\nabla} \times \vec{H} = \vec{J} = \frac{1}{\mu} \vec{\nabla} \times \vec{B}, \quad \vec{\nabla} \times \vec{E} + \partial_t \vec{B} = 0$$

$$\frac{1}{\mu\sigma} \vec{\nabla} \times \left(\vec{\nabla} \times \vec{B} \right) = \vec{\nabla} \times \vec{E} + \vec{\nabla} \times \left(\vec{v} \times \vec{B} \right)$$

$$\frac{1}{\mu\sigma} \left[\vec{\nabla} \left(\vec{\nabla} \cdot \vec{B} \right) - \nabla^2 \vec{B} \right] = -\partial_t \vec{B} + \vec{\nabla} \times \left(\vec{v} \times \vec{B} \right)$$

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- which gives, for the B-field

$$\partial_t \vec{B} = \vec{\nabla} \times \left(\vec{v} \times \vec{B} \right) + \frac{1}{\mu\sigma} \nabla^2 \vec{B}$$

where we can drop the 2nd term on the RHS for large σ

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- Hydrodynamics (ignoring gravity)

$$\begin{aligned} \partial_t \rho + \vec{\nabla} \cdot (\rho \, \vec{v}) &= 0 \\ \rho \, \partial_t \vec{v} + \rho \left(\vec{v} \cdot \vec{\nabla} \right) \vec{v} &= -\vec{\nabla} p - \frac{1}{\mu} \vec{B} \times \left(\vec{\nabla} \times \vec{B} \right) \end{aligned}$$

  with the continuity equation and force density from mechanical pressure ($\vec{\nabla} p$) and magnetic force ($\vec{J} \times \vec{B}$)

Magnetohydrodynamic waves

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- and the magnetic force given by

$$-\frac{1}{\mu} \vec{B} \times \left(\vec{\nabla} \times \vec{B}\right) = -\vec{\nabla} \left(\frac{1}{2\mu} \left|\vec{B}\right|^2\right) + \frac{1}{\mu} \left(\vec{B} \cdot \vec{\nabla}\right) \vec{B}$$

Magnetohydrodynamic waves

$$
\begin{aligned}
\partial_t \rho + \vec{\nabla} \cdot \left[\rho\, \vec{v}\right] &= 0 \\
\rho \left(\partial_t + \vec{v} \cdot \vec{\nabla}\right) \vec{v} &= -s^2\, \vec{\nabla} \rho \\
&\quad - \frac{1}{\mu} \vec{B} \times \left[\vec{\nabla} \times \vec{B}\right] \\
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- Consider small amplitude waves in a stationary fluid with constant equilibrium ρ_0 in a uniform static external $\vec{B}_0$

$$
\vec{B} = \vec{B}_0 + \vec{B}_1(\vec{x}, t), \quad \rho = \rho_0 + \rho_1(\vec{x}, t), \quad \vec{v} = \vec{v}_1(\vec{x}, t)
$$

Magnetohydrodynamic waves

$$\partial_t(\rho_0+\rho_1)+\vec{\nabla}\cdot\left[(\rho_0+\rho_1)\,\vec{v}_1\right]=0$$

$$(\rho_0+\rho_1)\left(\partial_t+\vec{v}_1\cdot\vec{\nabla}\right)\vec{v}_1=-s^2\,\vec{\nabla}(\rho_0+\rho_1)$$
$$-\frac{1}{\mu}(\vec{B}_0+\vec{B}_1)\times\left[\vec{\nabla}\times(\vec{B}_0+\vec{B}_1)\right]$$

$$\partial_t(\vec{B}_0+\vec{B}_1)=\vec{\nabla}\times\left[\vec{v}_1\times(\vec{B}_0+\vec{B}_1)\right]$$

- Consider small amplitude waves in a stationary fluid with constant equilibrium ρ_0 in a uniform static external $\vec{B}_0$

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Magnetohydrodynamic waves

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$$\partial_t(\vec{B}_0+\vec{B}_1)=\vec{\nabla}\times\left[\vec{v}_1\times(\vec{B}_0+\vec{B}_1)\right]$$

- Keeping only1st order terms in $\vec{B}_1,\rho_1,\vec{v}_1$

$$\partial_t\rho_1+\rho_0\vec{\nabla}\cdot\vec{v}_1=0$$

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$$\begin{aligned} \partial_t \rho_1 + \rho_0 \vec{\nabla} \cdot \vec{v}_1 &= 0 \\ \rho_0 \partial_t \vec{v}_1 + s^2 \vec{\nabla} \rho_1 + \frac{1}{\mu} \vec{B}_0 \times \left(\vec{\nabla} \times \vec{B}_1 \right) &= 0 \\ \partial_t \vec{B}_1 - \vec{\nabla} \times \left(\vec{v}_1 \times \vec{B}_0 \right) &= 0 \end{aligned}$$

- Manipulating and combining these

$$\begin{aligned} \partial_t \vec{\nabla} \rho_1 + \rho_0 \vec{\nabla} \left(\vec{\nabla} \cdot \vec{v}_1 \right) &= 0 \\ \rho_0 \partial_t^2 \vec{v}_1 + s^2 \partial_t \vec{\nabla} \rho_1 + \frac{1}{\mu} \vec{B}_0 \times \partial_t \left(\vec{\nabla} \times \vec{B}_1 \right) &= 0 \\ \partial_t \left(\vec{\nabla} \times \vec{B}_1 \right) - \vec{\nabla} \times \left[\vec{\nabla} \times \left(\vec{v}_1 \times \vec{B}_0 \right) \right] &= 0 \end{aligned}$$

Magnetohydrodynamic waves

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$$\partial_t \vec{\nabla} \rho_1 + \rho_0 \vec{\nabla} \left(\vec{\nabla} \cdot \vec{v}_1 \right) = 0$$

$$\rho_0 \partial_t^2 \vec{v}_1 + s^2 \partial_t \vec{\nabla} \rho_1 + \frac{1}{\mu} \vec{B}_0 \times \partial_t \left(\vec{\nabla} \times \vec{B}_1 \right) = 0$$

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- we get a wave equation

$$\partial_t^2 \vec{v}_1 - s^2 \vec{\nabla} \left(\vec{\nabla} \cdot \vec{v}_1 \right) + \vec{v}_A \times \vec{\nabla} \times \left[\vec{\nabla} \times (\vec{v}_1 \times \vec{v}_A) \right] = 0$$

Magnetohydrodynamic waves

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- where $\vec{v}_A = \vec{B}_0/\sqrt{\mu\,\rho_0}$ is called the *Alfvèn velocity*
- Looking for harmonic plane wave solutions

$$\vec{v}_1(\vec{x},t) = \vec{v}_1\, e^{i\left(\vec{k}\cdot\vec{x}-\omega\, t\right)}$$

$$-\omega^2 \vec{v}_1 + \left(s^2 + v_A^2\right)\left(\vec{k}\cdot\vec{v}_1\right)\vec{k}$$

$$+\left(\vec{v}_A\cdot\vec{k}\right)\left[\left(\vec{v}_A\cdot\vec{k}\right)\vec{v}_1 - \left(\vec{v}_A\cdot\vec{v}_1\right)\vec{k} - \left(\vec{k}\cdot\vec{v}_1\right)\vec{v}_A\right] = 0$$

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- For $\vec{k}\perp\vec{v}_A$, we get a longitudinal ($\vec{v}_1 = v_1\hat{\mathbf{k}}$) *magnetosonic* wave with phase velocity $u = \sqrt{s^2+v_A^2}$

$$-\omega^2\vec{v}_1 + \left(s^2+v_A^2\right)\left(\vec{k}\cdot\vec{v}_1\right)\vec{k} = 0 \rightarrow \left(\frac{\omega}{k}\right)^2 = s^2+v_A^2$$

Magnetohydrodynamic waves

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$$- \omega^2 \vec{v}_1 + \left(s^2 + v_A^2\right) \left(\vec{k} \cdot \vec{v}_1\right) \vec{k}$$

$$+ \left(\vec{v}_A \cdot \vec{k}\right) \left[\left(\vec{v}_A \cdot \vec{k}\right) \vec{v}_1 - \left(\vec{v}_A \cdot \vec{v}_1\right) \vec{k} - \left(\vec{k} \cdot \vec{v}_1\right) \vec{v}_A\right] = 0$$

- For $\vec{k} \parallel \vec{v}_A$

$$\left(k^2 v_A^2 - \omega^2\right) \vec{v}_1 + \left(\frac{s^2}{v_A^2} - 1\right) k^2 \left(\vec{v}_A \cdot \vec{v}_1\right) \vec{v}_A = 0$$

and there are 2 possibilities

Magnetohydrodynamic waves

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and there are 2 possibilities

- ordinary longitudinal wave ($\vec{v}_1 \parallel \vec{k} \parallel \vec{v}_A$) with phase velocity s and transverse wave ($\vec{v}_1 \cdot \vec{v}_A = 0$) with phase velocity $\vec{v}_A$

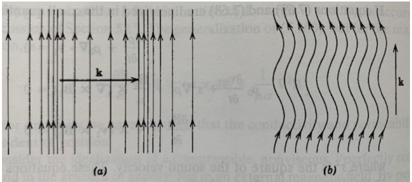

Figure 7.12 Magnetohydrodynamic waves.

Magnetohydrodynamic waves

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Corresponding magnetic field follows from

$$\partial_t \vec{B}_1 = \vec{\nabla} \times (\vec{v}_1 \times \vec{B}_0)$$

and is given by

$$\vec{B}_1 = \begin{cases} \frac{k}{\omega} v_1 B_0, & \vec{k} \perp \vec{B}_0 \\ \vec{0}, & \text{longitudinal } \vec{k} \parallel \vec{B}_0 \\ -\frac{k}{\omega} B_0 \vec{v}_1, & \text{transverse } \vec{k} \parallel \vec{B}_0 \end{cases}$$

Superposition and group velocity

- Worked with waves of single ω, k so far
- Real signal sources have finite frequency spread
- Linearity of Maxwell eqns allows superposition but some important details worth closer look
- In a dispersive medium, the wave is distorted
  - phase velocity varies with frequency, so different frequency components donot remain phase locked

$$\epsilon = \epsilon(\omega) \rightarrow v_p = \frac{\omega}{k} = v_p(\omega)$$

  - Group velocity

$$v_g = \frac{d\omega}{dk}$$

    may be very different from v_p and even indeterminate
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Superposition in 1D

- Scalar wave in one dimension $u(x,t)$ (can think of it as a component of the field, eg. E_y)
- Wave eqn
$$\partial_x^2 u - \frac{1}{v^2}\partial_t^2 u = 0$$
- General soln
$$u(x,t) = f(x - vt) + g(x + vt)$$
- Plane-wave solns
$$u(x,t) = a\, e^{ik(x-vt)} + b\, e^{-ik(x+vt)}$$
- Frequency-wavevector (dispersion) relation for medium
$$\omega = \omega(k) = v(k)\, k$$
- Even ($\omega(-k) = \omega(k)$) since it shouldn't depend on direction of propagation, and mostly smooth (except for "anomalous dispersion" near resonant absorption)

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Superposition in 1D

- (Spatial) Fourier decomposition of wave soln assuming real k, ω, no dissipation (note different normalization convention)

$$u(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} A(k)\, e^{i[kx-\omega(k)t]}\, dk$$

- If there is no dispersion,
  - plane wave components at different k have the same velocity v
  - wave shape preserved as it moves through the medium

Superposition in 1D

- (Spatial) Fourier decomposition of wave soln assuming real k, ω, no dissipation (note different normalization convention)

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$$\begin{aligned} \omega = vk \qquad & \rightarrow u(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} A(k)\, e^{ik(x-vt)}\, dk \\ u(x,0) = u_0(x) \qquad & u(x,t) = u_0(x-vt) \end{aligned}$$

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$$u(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} A(k) e^{i[kx-\omega(k)t]}\, dk$$

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- A pure harmonic function of space at $t = 0$ propagates as a monochromatic plane wave

$$u(x,0) = e^{ik_0 x} \rightarrow u(x,t) = e^{i[k_0 x - \omega(k_0)t]}$$

Propagation of a wavepacket

$$u(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} A(k)\, e^{i[kx-\omega(k)t]}\, dk$$

- For $A(k)$ sharply peaked at $k = k_0$ with width Δk, the spatial waveform is a pulse of width Δx with

$$\Delta x \, \Delta k \geq 1/2$$

- Expanding ω around k_0

$$\omega(k) = \omega_0 + v_g(k - k_0) + \ldots, \quad \omega_0 = \omega(k_0), \quad v_g = \left.\frac{d\omega}{dk}\right|_{k_0}$$

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$$u_0(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} A(k)\, e^{ikx}\, dk$$

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- but not meaningful in that case due to strong absorption and the smooth $n(\omega)$ approximation is not valid